Tutorial 2 - Linear and Circular ConvolutionLinear Convolution using DFT

Linear filtering is same as linear convolution.

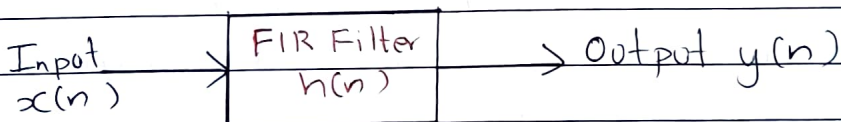
We know that linear convolution of $x(n)$ and $h(n)$ is given by,

$$y(n) = x(n) * h(n)$$

$$= \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

$$= \sum_{k=-\infty}^{\infty} h(k) x(n-k) \rightarrow (1)$$

Consider FIR filter having impulse response $h(n)$ as shown in Fig 1



Length 'L'

Length 'M'

Length $N = L + M - 1$

Fig 1 - FIR Filter



Here $x(n)$ = Input sequence having length 'L'

$$\therefore x(n) = \{0, 1, 2, \dots, L-1\}$$

$h(n)$ = Impulse response of filter having length 'M'

$$\therefore h(n) = \{0, 1, 2, \dots, M-1\}$$

Here the linear convolution of $x(n)$ and $h(n)$ produces the output sequence $y(n)$ and the length of $y(n)$ is

$$N = L + M - 1 \rightarrow (2)$$

Steps to Calculate Linear Filtering using DFT

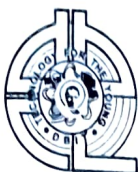
Step 1:- Calculate the value of N using $N = L + M - 1$.
Here 'L' represents number of samples in $x(n)$ and M represents number of samples in $h(n)$.

Step 2:- By adding zeros, make the length of $x(n)$ and $h(n)$ equal to N .

Step 3:- Calculate DFT of $x(n)$ that means $X(k)$

Step 4:- Calculate DFT of $h(n)$ that means $H(k)$

Step 5:- Multiply $X(k)$ and $H(k)$ to get



$$Y(k) = X(k) \cdot H(k)$$

Step 6:- Obtain IDFT of $Y(k)$ that means $y(n)$

Problem

1) Using DFT and IDFT method compute linear convolution ~~use~~ of two sequences

$$x(n) = \{2, 3, 5\}$$

$$h(n) = \{1, 3\}$$

Solⁿ Step 1:- Length of $x(n) = L = 3$
Length of $h(n) = 2$

$$\begin{aligned} \therefore N &= L + M - 1 \\ &= 3 + 2 - 1 \\ &= 4 \end{aligned}$$

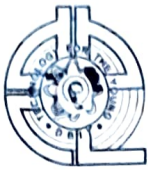
Step 2:- Make lengths of $x(n)$ and $h(n)$ equal to 4.

$$\therefore x(n) = \{2, 3, 5, 0\}$$

$$h(n) = \{1, 3, 0, 0\}$$

Step 3:- Calculate $X(k)$

$$X(k) = [W_4] \cdot x_N$$



$$X(k) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 5 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2+3+5+0 \\ 2-3j-5+0 \\ 2-3+5+0 \\ 2+3j-5+0 \end{bmatrix} = \begin{bmatrix} 8 \\ -3-3j \\ +4 \\ -3+3j \end{bmatrix}$$

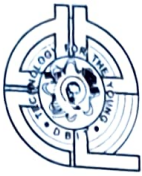
Step 4:- Calculate $H(k)$

$$H(k) = [W_4] h_N = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1+3+0+0 \\ 1-3j+0+0 \\ 1-3+0+0 \\ 1+3j+0+0 \end{bmatrix} = \begin{bmatrix} 4 \\ 1-3j \\ -2 \\ 1+3j \end{bmatrix}$$

Steps: Calculate $Y(k) = X(k) \cdot H(k)$

$$\therefore Y(k) = \{10, -3-3j, 4, -3+3j\} \cdot \{4, 1-3j, -2, 1+3j\} \\ = \{40, -12+6j, -8, -12-6j\}$$



$$\boxed{j^2 = -1}$$

$$\begin{aligned} (-3-3j)(1-3j) &= -3-3j+9j-9 \\ &= -12+6j \end{aligned}$$

$$\begin{aligned} (-3+3j)(1+3j) &= -3+3j+9j-9 \\ &= -12-6j \end{aligned}$$

Q.

Step 6 :- Calculate IDFT of $Y(k)$

$$\therefore y(n) = \frac{1}{4} [W_4^k] Y_N$$

$$= \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 40 \\ -12+6j \\ -8 \\ -12-6j \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 40-12+6j-8-12-6j \\ 40-12j-6+8+12j-6 \\ 40-12-6j-8+12+6j \\ 40+12j+6+8-12j+6 \end{bmatrix}$$

$$= \frac{1}{40} \begin{bmatrix} 8 \\ 36 \\ 56 \\ 60 \end{bmatrix} = \begin{bmatrix} 2 \\ 9 \\ 14 \\ 15 \end{bmatrix}$$

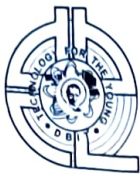
The linear convolution is $y[n] = \{2, 9, 14, 15\}$



Q2) ~~Q2)~~ Using DFT and IDFT method. Compute linear convolution of two sequences

$$x(n) = \{1, 2, 1\}$$

$$h(n) = \{1, 1\}$$



Circular Convolution

$$\text{If } x_1(n) \xleftrightarrow{\text{DFT}} X_1(k)$$

$$\text{And } x_2(n) \xleftrightarrow{\text{DFT}} X_2(k)$$

$$\text{then } x_1(n) \circledast x_2(n) \xleftrightarrow{\text{DFT}} X_1(k) \cdot X_2(k) = X_3(k)$$

Hence $\circledast$ is the symbol used for circular convolution

We find $x_3(n)$ for which DFT is $X_3(k)$,

From the Periodic Convolution

$$x_{3p}(n) = \sum_{m=0}^{N-1} x_{1p}(m) \cdot x_{2p}(n-m)$$

The subscript p stands for periodic.

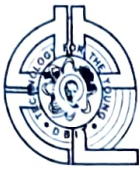
$$\text{i.e. } x_3(n)_N = \sum_{m=0}^{N-1} x_1(m)_N \cdot x_2(n-m)_N$$

for $0 \leq n \leq N-1$

$$x_3(n)_N = x_3(n)$$

$$x_1(m)_N = x_1(m)$$

$$\therefore x_3(n) = \sum_{m=0}^{N-1} x_1(m) \cdot x_2(n-m)_N \rightarrow (3)$$



Now RHS of eq (3) represents circular convolution of $x_1(n)$ and $x_2(n)$ which is represented by $\otimes$

$$x_3(n) = x_1(n) \otimes x_2(n)$$

$$\begin{aligned} \therefore \text{DFT}\{x_3(n)\} &= \text{DFT}\{x_1(n) \otimes x_2(n)\} \\ &= X_3(k) \\ &= X_1(k) \cdot X_2(k) \end{aligned}$$

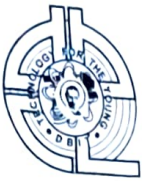
$$\therefore x_1(n) \otimes x_2(n) \xrightarrow{\text{DFT}} X_1(k) \cdot X_2(k)$$

Hence circular convolution reduces to multiplication while working with the DFT.

Matrix multiplication Method

In this method, we generate a circular matrix of $x_2(n)$ and multiply it by $x_1(n)$.

$$\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ \vdots \\ y(N-1) \end{bmatrix} = \begin{bmatrix} x_2(0) & x_2(N-1) & x_2(N-2) & \dots & x_2(1) \\ x_2(1) & x_2(0) & x_2(N-1) & \dots & x_2(2) \\ x_2(2) & x_2(1) & x_2(0) & \dots & x_2(3) \\ \vdots & \vdots & x_2(1) & \dots & \vdots \\ x_2(N-2) & \vdots & \vdots & \dots & \vdots \\ x_2(N-1) & x_2(N-2) & \vdots & \dots & x_2(0) \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_1(1) \\ x_1(2) \\ x_1(3) \\ x_1(4) \\ x_1(N-1) \end{bmatrix}$$



$y(0)$	$x_2(0)$	$x_2(N-1)$	$x_2(N-2)$	$\dots$	$x_2(1)$	$x_1(0)$
$y(1)$	$x_2(1)$	$x_2(0)$	$x_2(N-1)$	$\dots$	$x_2(2)$	$x_1(1)$
$y(2)$	$x_2(2)$	$x_2(1)$	$x_2(0)$	$\dots$	$x_2(3)$	$x_1(2)$
$y(3)$	$\vdots$	$\vdots$	$x_2(1)$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$x_2(N-2)$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$y(N-1)$	$x_2(N-1)$	$x_2(N-2)$	$\vdots$	$\vdots$	$x_2(0)$	$x_1(N-1)$

Note:- Since the next column of the circular matrix will be same as the first column we stop. The x_2 matrix is called a circular matrix.

Problem:-

- 1) Perform circular convolution on the given two sequences $x_1(n) = \{1, 2, 3, 4\}$
 $x_2(n) = \{4, 1, 1, 2\}$

Soln) We generate a circular matrix of $x_2(n)$ and multiply by $x_1(n)$

$$y(n) = x_1(n) \otimes x_2(n)$$



$$\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{bmatrix} = \begin{bmatrix} 4 & 2 & 1 & 1 \\ 1 & 4 & 2 & 1 \\ 1 & 1 & 4 & 2 \\ 2 & 1 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 4+4+3+4 \\ 1+6+6+4 \\ 1+2+12+8 \\ 2+2+3+16 \end{bmatrix}$$

$$= \begin{bmatrix} 15 \\ 17 \\ 23 \\ 23 \end{bmatrix}$$

(Q2) Perform circular convolution on the given two sequences

$$x(n) = \{1, 2, 3, 1\}$$

$$h(n) = \{4, 3, 2, 2\}$$